

RAG ASSIGNMENTS

Assignment 1: PDF Chatbot using Managed Pinecone or QDrant vector DB

Instead of using a local chromadb, use a Managed Service vector database.

Several of them offer a free tier, such as pinecone, QDrant.

Explore the difference between them, their capabilities

Replace the vector db in assignment 1) with a managed service vector db

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Working HF URL (in a text file inside RAG folder under your shared assignments folder)

Assignment 2: Re-ranking to improve retrieval results

Tasks:

1. Load 100 Wikipedia documents
2. Split them into 500-character chunks
3. Create embeddings using Bedrock's Titan model
4. Store them in a local ChromaDB instance
5. Perform retrieval **with** and **without re-ranking**
You may use Amazon Bedrock Claude 3.5 LLM as re-ranker or Cohere (which is a popular one <https://cohere.com/rerank>)
6. Use Amazon Bedrock Claude 3.5 as the LLM
7. Run several queries showing the difference between baseline and reranked results

Tips

1. You may use the following to load wikipedia documents:

```
loader = HuggingFaceDatasetLoader(  
    dataset_name="Cohere/wikipedia-22-12-simple-embeddings",  
    page_content_column="text",  
    name="train",  
    load_max_docs=100 # Using just 100 documents for simplicity  
)
```

```
documents = loader.load()
```

2. You may use following queries to compare:

Compare retrieval methods with sample queries

```
queries = [
```

```
    "What are the main causes of climate change?",
```

```
    "How does quantum computing work?",
```

```
    "What were the social impacts of the industrial revolution?"
```

```
]
```

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